LEAN HOG FUTURES (HE)

Spread & Curve Structure — Fundamental Reference

Production Cycles, Slaughter Data, Demand, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of CME Lean Hog futures (ticker: HE), with particular focus on the forces that shape the forward curve and the spread structure between contract months. Lean Hog is a cash-settled contract with no physical delivery obligation, yet its pricing is tightly anchored to the physical pork market through the CME Lean Hog Index. The market is characterised by pronounced seasonality in both supply and demand, relatively short production cycles compared to cattle, and a complex interaction between domestic pork consumption, export trade flows (particularly to China and Mexico), and the broader livestock complex. Understanding each of these layers is the foundation of spread and butterfly trading on the Lean Hog curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement

Lean Hog futures are cash-settled, meaning no physical delivery of hogs takes place. The contract settles against the CME Lean Hog Index, a two-day weighted average of the USDA National Daily Direct Afternoon Hog report. This cash-settlement mechanism creates a predictable convergence dynamic as contracts approach expiry, and spreads between contract months reflect expectations about the Index level at each settlement date.

Parameter	Detail
Exchange	CME Group (Chicago Mercantile Exchange)
Ticker	HE (front month)
Contract size	40,000 pounds of lean hog carcass
Quotation	US cents per pound (¢/lb); minimum tick = 0.025¢/lb = $10.00 per contract
Contract months	February (G), April (J), May (K), June (M), July (N), August (Q), October (V), December (Z) — 8 months per year
Settlement type	Cash settlement against the CME Lean Hog Index (2-day weighted average of USDA National Daily Direct Afternoon Hog report prices)
Last trading day	10th business day of the contract month
Trading hours	CME Globex: Sunday–Friday 08:30–13:05 CT (pit hours); electronic: Sunday–Friday 07:00–13:05 CT and 14:00–07:00 CT next day
Open interest	Concentrated in the front 3–4 contracts; spreads and flies tradeable through approximately C6–C7 with declining liquidity beyond
Price limit	$4.50/cwt initial daily limit; expandable under CME rules

1.1 The CME Lean Hog Index

Because settlement is cash-based against the CME Lean Hog Index, understanding how the Index is constructed is important for interpreting basis behaviour and spread convergence:

- The Index is a 2-day weighted average of negotiated and formula-based pork carcass prices reported daily by USDA Agricultural Marketing Service (AMS).
- The Index reflects carcass value, not live hog prices. Lean Hog futures therefore price the expected carcass market at the time of contract expiry, not live weight or liveweight equivalents.
- The "basis" (CME Lean Hog Index minus nearest futures price) converges to zero at expiry. The direction and pace of this convergence is a signal about whether the physical cash market is stronger or weaker than what futures have priced.
- The Index is published daily by CME and is available at cmegroup.com. It is the primary reference for front-contract fair-value assessment.

1.2 Contract Month Structure and the Crop Year Analogy

The eight contract months are not evenly distributed — they reflect the seasonal rhythm of hog production and slaughter. The absence of certain months (January, March, September, November) creates natural "gaps" in the curve that spread traders must account for when comparing adjacent contract spreads. The February and April contracts cover the seasonally slower first quarter; the May through August contracts cover the spring-to-summer production ramp; October and December bracket the autumn slaughter peak.

2. The Hog Production Cycle

The biological production cycle of hogs is the single most important structural feature of the Lean Hog market. Unlike crude oil or financial assets, hog supply is determined by breeding decisions made months in advance and cannot be quickly altered once the production pipeline is set. Understanding this cycle allows traders to anticipate supply changes with meaningful lead time.

2.1 The Production Timeline

Stage	Duration	Key Event	Futures Implication
Gestation	114 days (~3.8 months)	Sow bred; litter size typically 10–14 piglets	Breeding data today = farrowing in ~4 months = market hogs in ~10 months
Lactation / weaning	3–4 weeks	Piglets weaned at 21–28 days	Pig crop data from USDA Hogs & Pigs report is the key lagged signal
Nursery phase	6–8 weeks	Weaned pigs grow from ~15 lb to ~50 lb	—
Grow-finish phase	16–20 weeks	Hogs grow from ~50 lb to ~280 lb market weight	Key phase — feed cost economics drive producer decisions
Total birth-to-market	~5.5–6 months	Market hog ready for slaughter at ~280 lb live weight (~200 lb carcass)	Breeding decisions today = market supply ~6 months forward

The approximately 6-month birth-to-market timeline means that the USDA Hogs & Pigs quarterly report — which measures breeding inventory and farrowing intentions — is a leading indicator of market hog supply approximately 5-7 months forward. This is the most important single piece of data for mid-curve spread positioning.

2.2 The Breeding Herd and Sow Inventory

The size of the breeding herd (sow inventory) is the fundamental long-run supply variable. Changes to the breeding herd take approximately 10–12 months to fully manifest in market hog supply:

- **Sow herd expansion**: producers add gilts (young females) to the breeding herd when profitability is expected to be positive. A gilt added today will not produce her first litter for approximately 7–8 months (gilt development: 5–6 months to breeding weight, then gestation of 3.8 months). Her offspring will reach market a further 5.5–6 months later — approximately 13–14 months from the gilt-retention decision to additional market hogs.
- **Sow herd contraction**: producers remove sows from the breeding herd (early slaughter) when feed costs are high, hog prices are low, or disease reduces productivity. Sow slaughter data is reported weekly by USDA and is a leading indicator of future production tightening.
- **Herd turnover**: commercial sows typically remain in the breeding herd for 4–6 parity cycles (litters) before being culled. Average culling age is approximately 2–2.5 years. This creates a relatively stable base turnover of sow slaughter against which changes in the rate of culling are measured.

2.3 Litter Size and Pig Crop Productivity

- Average litter size has increased significantly over time due to genetic selection and management improvements: from approximately 8–9 pigs per litter in the 1990s to 10–12 in modern commercial operations. Rising litter size means the same sow inventory produces more market hogs per year, creating a structural supply-growth bias that is separate from changes in sow numbers.

- Pre-weaning mortality (typically 10–15%) and grow-finish mortality (typically 2–4%) affect the conversion from pigs farrowed to market hogs delivered. Disease outbreaks that raise mortality rates (PED, PRRS — see Section 5) create unexpected production shortfalls relative to what the breeding herd size would imply.
- Average market weight has also increased over time: hogs now typically reach market at 280–295 lb live weight vs 240–250 lb in earlier decades. Higher average weights increase carcass pounds produced per head slaughtered, so the relationship between head-count slaughter and total pork output must account for weight trends.

2.4 Seasonality of Production

Hog production has inherent seasonal rhythms, though modern confinement operations have reduced (but not eliminated) seasonality compared to outdoor production systems:

- **Farrowing seasonality**: in commercial confinement operations, farrowing is relatively consistent year-round. However, heat stress in summer reduces sow conception rates and litter sizes, creating a predictable dip in piglet production in Q3 that translates to a dip in market hog availability approximately 5–6 months later (Q4–Q1).
- **Market weight seasonality**: hogs gain weight faster in cooler temperatures. Fall and winter hogs typically come to market at heavier weights, increasing carcass output per head. Summer heat stress slows weight gain, reducing per-head output.
- **Slaughter seasonality**: slaughter runs are typically highest in autumn (October–December) as fall-marketed hogs hit the system, and lowest in late winter and early spring. This seasonal supply pattern directly creates recurring spread opportunities between the summer and autumn contract months.

3. Supply Data Sources

3.1 USDA Hogs & Pigs Report

Published quarterly (approximately March, June, September, December), this is the single most important report for mid-to-back curve Lean Hog spreads. It surveys the US hog industry and reports:

- **All hogs and pigs inventory**: total count of hogs on US farms. The headline number. Year-over-year comparison is the primary market signal.
- **Breeding herd inventory (sows + boars)**: the structural supply variable. A rising breeding herd is a lagged bearish signal for market hog supply 5–7 months forward. A falling breeding herd is bullish.
- **Market hog inventory by weight category**: hogs are categorised by live weight (under 50 lb, 50–119 lb, 120–179 lb, 180 lb and over). The distribution across weight categories provides a "pipeline view" of how many hogs are at each stage of grow-finish and when they will reach slaughter weight. This is a 2–4 month forward supply visibility tool.
- **Farrowing intentions**: producer intentions to farrow sows in the next two quarters. A leading indicator (subject to revision) for pig crop size 5–7 months forward.
- **Pig crop (pigs weaned)**: actual number of pigs farrowed and weaned in the prior quarter. The most accurate lagged supply signal.

The Hogs & Pigs report typically causes immediate repricing of C3–C6 spreads on its release date. The key variables are: breeding herd vs year-ago (structural signal), and the weight-category distribution vs prior year and vs market expectations (near-term supply pipeline).

3.2 USDA Weekly Slaughter Data

The USDA reports federally inspected (FI) hog slaughter weekly (Monday estimate, Friday final). This is the highest-frequency supply data point and the primary driver of front-spread movements:

- **Weekly FI slaughter (head count)**: total hogs slaughtered under federal inspection. Year-over-year and vs analyst consensus are the key comparisons.
- **Average live weight**: reported weekly. Above-trend weights signal hogs backing up in the system (supply pressure building). Below-trend weights signal current-demand absorption is keeping pace with supply.
- **Average carcass weight**: the most directly relevant metric for carcass market pricing. High carcass weights increase total pork output even without head-count growth.
- **Sow slaughter**: reported separately. An elevated sow slaughter rate is a leading indicator of breeding herd contraction — bullish for C3–C6 contracts as it signals future production tightening.
- **Barrow and gilt slaughter vs sow**: the ratio of market hog slaughter to sow slaughter provides a continuous read on whether the breeding herd is expanding (low sow slaughter) or contracting (high sow slaughter).

3.3 USDA Cold Storage Report

- Published monthly. Reports pork in commercial cold storage (frozen pork, broken out by cut: belly, ham, loin, shoulder, other).
- Rising cold storage signals that supply is exceeding near-term demand — bearish for the front curve. Falling storage signals demand absorption is keeping pace or exceeding supply.
- Pork belly cold storage is particularly watched: pork belly prices are highly volatile and a large belly surplus can depress cutout values even when other cuts are tight.
- Cold storage data for pork interacts with seasonal demand: stocks building into Q1 (slow retail demand period) are more bearish than the same build heading into Q2–Q3 (when grilling season demand recovers).

3.4 USDA Pork Carcass Cutout Value

The USDA AMS publishes daily pork cutout values — the composite value of a pork carcass based on primal cut prices. This is the demand-side anchor for the physical market:

- **Pork cutout value ($/cwt carcass)**: the weighted average value of all primal cuts (ham, loin, belly, shoulder, rib, trim). The cutout is the most important daily fundamental signal for the front Lean Hog contract.
- **Individual cut values**: ham, loin, and belly each have distinct seasonal demand patterns. Ham demand peaks around Easter and Thanksgiving/Christmas. Belly demand peaks in summer (bacon and grilling). Loin demand is relatively stable year-round.
- **Packer margin**: the spread between the cutout value and the live hog acquisition cost. Wide packer margins incentivise packers to maximise slaughter runs. Compressed margins lead to run reductions, which slow the rate of hog marketings.

4. Demand Fundamentals

4.1 Domestic Pork Consumption

The US is the world's third-largest pork consumer after China and the EU. Domestic per capita pork consumption is approximately 50–52 lb per year. Key domestic demand drivers:

- **Retail demand**: supermarket pork sales are the baseline demand channel. Retail demand for pork is relatively price-inelastic in the short run but responds to protein substitution at the margin (consumers shift between pork, beef, and chicken based on relative prices). Chicken is pork's closest substitute in the US retail market.
- **Foodservice demand**: restaurants, fast food, and institutional buyers account for approximately 40–45% of total pork demand. Foodservice demand is more cyclical than retail and responds to consumer confidence, restaurant traffic, and economic conditions.
- **Seasonal demand patterns**: ham demand spikes at Easter (April) and November–December (Thanksgiving/Christmas). Bacon and rib demand peaks in summer (grilling season, May–August). These seasonal demand patterns interact with supply seasonality to create the characteristic shape of Lean Hog spreads across the year.
- **Competing protein prices**: chicken (broiler) prices are the most important competitive reference. When broiler prices are low relative to pork, some consumer and foodservice demand shifts away from pork. Beef prices also matter for higher-income consumer substitution. The relative price of competing proteins is tracked via USDA retail price surveys.

4.2 Export Demand — The Most Volatile Demand Variable

US pork exports have grown substantially over the past two decades and now represent approximately 25–30% of total US pork production. Export demand is the most volatile demand component and the primary source of unexpected surprises in the Lean Hog market:

Export Destination	Share of US Pork Exports	Key Products	Primary Risk Factor	Curve Impact
Mexico	~30%	Ham, shoulder, offal	Peso-USD exchange rate, trade policy, domestic protein substitution	Consistent baseline; peso weakness reduces Mexican import purchasing power
China / Hong Kong	~20–25%	Offal, head/feet, other cuts preferred in China	African Swine Fever (ASF) outbreaks, trade tariffs (Section 301), diplomatic relations	Largest swing factor; ASF-driven demand surge 2019–2020 was historic
Japan	~15%	Loin, belly, ham — high-value cuts	Yen-USD rate, CPTPP competition from Canada/EU	Stable, price-sensitive premium buyer
South Korea	~8%	Loin, shoulder, offal	FTA trade dynamics, domestic ASF episodes	Moderate; grows with Korean income levels
Canada	~5%	Various	CUSMA (USMCA) trade framework, integrated North American market	Relatively stable, integrated supply chain
Other	~17–20%	Various	Trade policy, regional disease outbreaks	Diversified, no single dominant risk

4.3 China and African Swine Fever — The Dominant Export Demand Shock

The 2018–2019 African Swine Fever (ASF) outbreak in China destroyed an estimated 40–50% of China's hog herd (approximately 200–300 million animals), the largest livestock disease event in recorded history. The consequences for global pork trade were profound and provide a framework for understanding how disease-driven demand shocks propagate through the Lean Hog forward curve:

- Chinese pork imports increased from approximately 1.2 million tonnes (2018) to over 4.5 million tonnes (2020) as domestic production collapsed. US pork exports to China surged, pulling domestic supply away from the US market and supporting prices strongly.
- The demand shock was concentrated in offal and lower-value cuts (head meat, feet, stomachs) preferred in Chinese cuisine — cuts that have limited US domestic demand and low standalone value. Their export value directly added to the pork carcass cutout value, effectively providing a subsidy to US packer margins.
- Chinese herd recovery (2021–2023) was faster than initially expected, driven by large-scale industrial production. As Chinese domestic production recovered, import demand fell sharply, removing the support that had been embedded in the mid-curve.
- ASF remains endemic in many Asian countries and recurs in previously cleared areas. Any new major ASF outbreak in China or Southeast Asia is a potential upside demand shock for US pork exports that would steepen backwardation in the front and mid-curve.

The US-China trade relationship (Section 301 tariffs, retaliatory tariffs) adds a policy risk layer to the ASF demand variable. Chinese retaliatory tariffs on US pork (imposed 2018) reduced competitiveness of US product vs EU and Canadian pork in the Chinese market, even as ASF drove total import demand higher.

4.4 USDA Weekly Export Data

- **Weekly Export Sales Report (ESR)**: USDA FAS publishes weekly data on net US pork export sales (new bookings) and shipments. Strong weekly sales — particularly to China or Mexico — are an immediate bullish signal for front and near-term contracts.
- **Monthly export data (PortStats)**: detailed monthly breakdown by destination country and product type. More accurate but lagged by 4–6 weeks.
- **USDA WASDE (World Agricultural Supply and Demand Estimates)**: monthly report updating USDA's projections for US pork production, consumption, and trade. Revisions to export forecasts are market-moving for C3–C6 contracts.

5. Disease Risk — The Tail Event in Lean Hog Markets

Disease is a uniquely important fundamental factor in hog markets because it can simultaneously destroy supply and (if it spreads globally) create demand shocks. Several diseases have documented history of materially moving the Lean Hog forward curve.

5.1 Porcine Epidemic Diarrhea Virus (PEDv)

PEDv emerged in the US in May 2013 and spread rapidly through the US hog herd during 2013–2014. It is highly fatal in neonatal piglets (mortality rates approaching 100% in some herds) but has low mortality in older hogs.

- PEDv killed an estimated 8–10 million piglets in the US during the 2013–2014 outbreak, reducing the 2014 pig crop by approximately 7–10% and causing a sharp reduction in market hog supply 5–6 months later (autumn 2014).
- The market impact: Lean Hog futures rallied to record highs in spring-summer 2014, with nearby contracts trading above $1.30/lb — more than double the prices seen two years prior. Front spreads went into extreme backwardation.
- PEDv is now endemic in the US hog population at low levels. Cold and wet weather conditions in winter facilitate virus transmission, creating an annual winter PEDv risk season. Outbreaks are monitored via USDA APHIS disease surveillance.

5.2 Porcine Reproductive and Respiratory Syndrome (PRRS)

- PRRS is an endemic viral disease causing reproductive failure in sows (abortions, stillbirths, smaller litters) and respiratory disease in growing pigs. It is the most economically costly swine disease in the US, estimated to cost the industry $600–700 million per year in losses.
- PRRS affects both supply (reduced litter sizes, increased mortality) and the timing of slaughter (slower weight gain in affected herds). Seasonal PRRS peaks typically occur in autumn and winter.
- New PRRS strains (Lineage 1C "Rosalia" and related variants) have periodically caused elevated disease burden beyond the endemic baseline. These events are tracked via MSHMP (Morrison Swine Health Monitoring Project) reporting.

5.3 African Swine Fever (ASF)

ASF has not yet been detected in the continental United States as of the document date, though it is present in the Caribbean (Dominican Republic, Haiti). A US ASF detection would be the most severe market disruption scenario:

- The US hog industry would face immediate export market closures from all major trading partners upon ASF detection (standard OIE/WOAH protocols). This would cause a simultaneous collapse in export demand and a domestic supply glut — extremely bearish for all contracts.
- Conversely, ASF outbreaks in major competing pork-producing regions (EU — Poland, Germany, Italy have had outbreaks) can reduce competing export supply and support US export demand, which is bullish for US Lean Hog mid-curve contracts.
- USDA APHIS maintains a national ASF detection and response plan. Market participants monitor USDA and OIE/WOAH disease notifications continuously. Any confirmed ASF in North America would cause immediate and extreme price dislocations across the curve.

5.4 Foot and Mouth Disease (FMD)

- FMD affects cloven-hoofed animals including hogs. The US has been FMD-free since 1929. An FMD detection would trigger immediate export bans from virtually all trading partners — similar scenario to ASF on the export side.

- USDA APHIS, alongside the Department of Homeland Security, runs ongoing FMD surveillance. The Plum Island Animal Disease Center (and its successor, the National Bio and Agro-Defense Facility in Manhattan, Kansas) is the primary US facility for FMD research and response.

5.5 Senecavirus A (SVA) and Other Emerging Diseases

- Senecavirus A causes vesicular disease in pigs that resembles FMD, creating diagnostic challenges and temporary market uncertainty on detection until FMD is ruled out. Not economically severe in itself but creates news-driven volatility.
- Mycoplasma hyopneumoniae (Mycoplasmal Pneumonia) is endemic and causes chronic respiratory disease and slow growth, contributing to industry-wide productivity drag rather than acute events.

6. Seasonal Structure of the Lean Hog Curve

Lean Hog is one of the most seasonally structured commodity markets. The interaction of seasonal supply (production cycle) and seasonal demand (consumption patterns, export timing) creates a highly repetitive curve shape across years, though the magnitude varies with the underlying supply/demand balance. Understanding the seasonal baseline is the prerequisite for identifying when the current year deviates from it.

Period	Supply Dynamics	Demand Dynamics	Typical Curve Behaviour	Key Contracts
Jan–Feb	Post-holiday slaughter slowdown; cold weather may slow weight gain; lowest seasonal supply	Weakest retail demand; foodservice slow; low export pace	Deferred (Apr–May) typically at premium to Feb; backwardation uncommon	Feb (G), Apr (J)
Mar–Apr	Spring flush of market hogs — autumn-farrowed hogs reach market weight; supply increases	Easter ham demand peak; grilling season approaches; export pace picks up	Apr often at discount to May if spring supply flush outpaces demand recovery	Apr (J), May (K)
May–Jun	Strong supply — spring-farrowed hogs growing; slaughter runs at elevated seasonal pace	Memorial Day grilling season; bacon and rib demand strengthens; foodservice picks up	Summer contracts (Jun–Aug) can trade at premium if demand exceeds supply flush	May (K), Jun (M)
Jul–Aug	Heat stress slows weight gain; summer supply dip; hogs back up if packers reduce runs during heat	Peak grilling demand; strong retail and foodservice; July 4th and summer BBQ season	Summer backwardation common if heat stress tightens supply; Aug often firm	Jul (N), Aug (Q)
Sep–Oct	Autumn flush — summer-farrowed hogs reaching market; largest seasonal supply build begins	Post-summer demand normalisation; school season dampens foodservice somewhat	Oct typically weakens relative to summer months as supply surge approaches	Oct (V)
Nov–Dec	Peak seasonal slaughter; largest weekly kill numbers of the year; heavy carcass weights as cool weather aids gain	Holiday ham demand (Thanksgiving, Christmas); offsetting demand support	Dec often at discount to following Feb as holiday ham demand must absorb large kill	Dec (Z), Feb (G)

6.1 The Summer Premium — A Structural Feature

In most years, the May–August contracts trade at a premium to the surrounding winter contracts (February and October–December). This summer premium reflects:

- Heat stress reducing summer supply (lighter weights, potential packer run reductions during extreme heat events).
- Peak grilling-season demand for bacon, ribs, and fresh pork cuts.
- Strong export demand from Japan and Korea for high-value cuts during summer.

The magnitude of the summer premium varies year-to-year with the severity of heat stress, the level of cold storage heading into summer, and the strength of export demand. When cold storage is high and export demand is weak, the summer premium compresses or disappears.

6.2 The Autumn Discount — When Supply Peaks

- The October and December contracts often trade at a discount to the preceding summer months because the market anticipates the seasonal peak in slaughter runs and the heaviest carcass weights of the year.

- Holiday ham demand (November–December) provides a partial offset to this seasonal supply pressure. When holiday demand is strong and cold storage of hams is low, the autumn discount is smaller or eliminates.
- The October contract specifically captures the transition from peak supply to beginning of the holiday demand period. Its spread relationship to both August and December is one of the most watched flies in the Lean Hog market.

7. Feed Costs and Production Economics

Hog production profitability is determined by the margin between hog prices (revenue) and feed costs (the primary variable cost). Feed represents approximately 60–70% of the total cost of production in commercial hog operations. Corn and soybean meal are the two primary feed ingredients.

7.1 The Hog-Corn Ratio

The hog-corn ratio (bushels of corn equal in value to 100 lb of live hog) is the traditional profitability indicator for hog producers. It measures how many bushels of corn a producer can buy with the revenue from selling 100 lb of live weight hogs:

- Historical breakeven: approximately 16–20 bushels of corn per 100 lb of live hogs (varies with non-feed production costs). Above breakeven = profitable to produce; below = unprofitable.
- When the hog-corn ratio is persistently below breakeven, producers reduce breeding herd size, which reduces pig crops 5–7 months forward — a bullish signal for deferred contracts.
- When the ratio is persistently above breakeven, producers expand breeding herds — a bearish lagged signal for deferred supply.
- The ratio is less precise today than historically because soybean meal costs have become a larger input component and non-feed costs (labour, energy, capital) have risen. More sophisticated margin models use a blend of corn price, soybean meal price, and fixed overhead.

7.2 Corn Prices and Their Impact on Supply

- Corn is the primary energy source in hog feed rations, comprising 60–70% of the diet by weight. CBOT Corn futures (ZC) are the reference price. High corn prices directly compress producer margins and eventually reduce production incentives.
- The lag between high corn prices and supply reduction is approximately 6–12 months (time for producers to adjust farrowing intentions, reduce sow inventory, and see the effect in market hog supply).
- The 2012 US drought drove corn prices to record highs ($8+/bushel), compressing hog margins severely and triggering a breeding herd reduction that contributed to tighter hog supplies in 2013–2014 (compounded subsequently by PEDv).

7.3 Soybean Meal and the Protein Premium

- Soybean meal (SBM) provides the protein component of hog rations, comprising 15–25% of the diet. CBOT Soybean Meal futures (ZM) are the reference price.
- Soybean meal prices are influenced by global soybean supply (South American and US crops), crush economics, and competition from alternative protein sources (DDGS, field peas).
- When both corn and soybean meal are simultaneously expensive (as occurred in 2021–2022), the margin squeeze on hog producers is particularly severe and the signal for future supply reduction is stronger.

7.4 Feed Cost Spread Relationships

The relationship between feed commodity futures (corn, soybean meal) and Lean Hog futures creates inter-commodity spread opportunities that are tracked by livestock-feed traders:

- **Hog crush margin**: buy corn and soybean meal (the inputs), sell Lean Hog futures (the output). This is the producer's economic position replicated in futures. Wide crush margin = bullish production incentive (potential supply increase). Narrow/negative crush = bearish production incentive (potential supply reduction).

- **DDGS (distillers dried grains with solubles)**: a byproduct of ethanol production from corn. Available as a partial corn substitute in hog rations. High ethanol production = more DDGS available = lower effective feed costs. The price relationship between corn and ethanol affects DDGS availability and hog feed economics.

8. Packer Dynamics, Capacity, and the Cash Market

8.1 US Pork Packing Industry Structure

The US pork packing industry is highly concentrated. Four companies — Smithfield Foods, Tyson Foods, JBS USA, and Seaboard Foods — control approximately 65–70% of total US hog slaughter capacity. This concentration means that individual packer decisions (run rates, maintenance schedules, labour actions) can have material effects on weekly slaughter data and near-term spread pricing.

- **Packing plant kill capacity**: approximately 480,000–500,000 head per day federally inspected capacity in the US. Weekly slaughter is typically 460,000–490,000 head when plants are running at full capacity. Slaughter significantly below this range indicates either packer-side reduction (maintenance, labour, margin) or supply-side tightness.
- **Plant maintenance shutdowns**: packing plants take scheduled annual maintenance typically in late January and again in summer. These reduce weekly kill capacity for 1–3 weeks and create predictable dips in FI slaughter numbers.
- **Labour disruptions**: meat packing is labour-intensive. Worker illness (COVID-19 caused significant plant closures in 2020), labour shortages, or strikes can rapidly reduce kill capacity and create hog backups on farms — a near-term bearish signal for live hog prices as supply piles up, but potentially bullish once the backup clears.

8.2 The Negotiated Cash Market

- **Daily negotiated hog purchases**: a portion of US hog production is sold daily in negotiated cash transactions (as opposed to formula-based contracts or packer-owned hogs). The negotiated price is the input to the CME Lean Hog Index.
- **Formula-based and contract purchases**: the majority of US hog production is sold under formula or long-term contracts, typically priced off the CME Lean Hog Index or the cutout value. This structure creates a self-referencing relationship between futures prices, cash prices, and producer revenues.
- **Packer-owned hogs**: large packers own a significant share of their own hog supply (vertical integration). Smithfield Foods, for example, owns a large portion of its supply chain. Packer-owned hogs do not transact in the cash market and therefore do not affect the negotiated price used in the Index.
- **Thin cash market concern**: the declining volume of negotiated cash transactions (as more production moves to formula contracts and vertical integration) has raised questions about the representativeness and reliability of the CME Lean Hog Index. USDA and CME have periodically reviewed the Index methodology in response to these concerns.

9. Macro, Financial, and Cross-Market Drivers

9.1 Currency Effects on Export Demand

- **USD vs MXN (Mexican peso)**: Mexico is the largest single export destination for US pork. When the peso weakens against the dollar, US pork becomes more expensive in peso terms, reducing Mexican import purchasing power. This is a persistent and documentable driver of monthly export data.
- **USD vs JPY (Japanese yen)**: Japan is a major buyer of high-value US pork cuts (loin, belly). Yen weakness reduces Japanese purchasing power for USD-denominated imports.
- **USD vs CNY (Chinese yuan)**: China's pork import demand is priced in USD. A weaker yuan reduces Chinese import competitiveness at the margin, though ASF-driven demand can overwhelm currency effects when outbreaks are severe.

9.2 Trade Policy Risk

- **Section 301 tariffs (China)**: China imposed retaliatory tariffs of 25% on US pork in response to US tariffs in 2018. These tariffs redirected Chinese pork import demand toward EU, Canadian, and Brazilian suppliers. Any modification to these tariffs (reduction or escalation) would immediately affect US pork export competitiveness in China.
- **USMCA / CUSMA**: the US-Mexico-Canada trade agreement governs trade with the two largest US agricultural trading partners. Stable framework; changes would be highly market-moving given Mexico's importance as a pork export destination.
- **Japan trade agreements**: the US-Japan Trade Agreement (2020) and subsequent negotiations have affected Japanese tariff rates on US pork. Competition from CPTPP members (Canada, EU association agreement) affects the price competitiveness of US pork in Japan.
- **Country of origin labelling (COOL)**: US labelling rules affect the competitiveness of Canadian hogs processed in the US for export. Policy changes can alter the economics of Canadian hog imports for US packers.

9.3 Competing Proteins

- **Broiler (chicken) production**: the US broiler industry is the largest and most efficient protein producer globally. Broiler production grows at approximately 1–2% per year. Low broiler prices compete directly with pork in the retail and foodservice channels. The USDA Broiler Hatchery report (weekly) and NASS Poultry Production reports track broiler supply.
- **Beef production cycle**: the beef cattle cycle (7–10 years) creates periods of high and low beef supply that affect consumer protein choice. During periods of high beef supplies and low beef prices, consumer protein spending may shift partially toward beef, reducing pork demand at the margin.
- **Alternative proteins**: plant-based protein products represent a small but growing share of the protein market. Their penetration affects primarily the processed meat segment (hot dogs, sausage), not the fresh pork cut market.

9.4 Speculative Positioning — COT Data

- **Managed money net position**: hedge funds and CTAs in Lean Hog futures. As with other livestock markets, extreme net long or net short positioning signals crowding risk. The Lean Hog managed money position has historically shown strong seasonal patterns — building long positions heading into the summer premium period and liquidating in autumn.
- **Commercial hedging**: packers hedge their procurement costs in Lean Hog futures (buying futures to lock in hog costs). Producers may sell futures to hedge forward production. The commercial net position reflects the balance of producer vs packer hedging at any point in time.

10. Front vs Back — Key Differences and Practical Framework

Dimension	Front of Curve (C1–C3)	Back of Curve (C4–C7)
Primary driver	Weekly FI slaughter, daily pork cutout, cash hog price vs Index	USDA Hogs & Pigs breeding herd + pig crop data
Data frequency	Daily (cutout) to weekly (slaughter, weights)	Quarterly (Hogs & Pigs) + monthly (WASDE, Cold Storage)
Info-to-price lag	Hours to 3 days	Weeks to months
Holding period	Days to 3 weeks	3 weeks to 3 months
Liquidity	Good — 20–60 lots workable in most conditions	Thin — 5–15 lots typical; wider bid-ask
Dominant participants	Packers (hedging procurement), index funds, retail speculators	Producer hedgers (selling forward), managed money, commercials
Seasonal character	Immediate: captures real-time supply-demand imbalance	Structural: captures expected seasonal pattern 3–6 months forward
Disease sensitivity	Immediate: PEDv/PRRS news moves front within hours	Delayed: disease effect priced as front moves and extends to back
Export sensitivity	Weekly ESR data moves front 1–3 days	WASDE export revisions move back curve on release day
Main risk	Surprise in weekly slaughter or weights; disease news	Hogs & Pigs report surprise; unexpected feed cost shift; disease

11. Key Data Release Schedule

Report	Publisher	Frequency	Release Timing	Primary Curve Impact
USDA Hogs & Pigs	USDA NASS	Quarterly	Late Mar/Jun/Sep/Dec	C3–C7 — most important structural report
Weekly FI Slaughter (estimate)	USDA NASS	Weekly	Monday ~3:00 PM ET	C1–C2 — most immediate weekly supply signal
Weekly FI Slaughter (final)	USDA NASS	Weekly	Friday ~3:00 PM ET	C1–C2 — confirms Monday estimate
Pork Carcass Cutout Value	USDA AMS	Daily	11:00 AM / 2:00 PM ET	C1 — primary daily demand signal
USDA Cold Storage Report	USDA NASS	Monthly	Last business day of month	C1–C4 — pork inventory; belly stocks watched closely
USDA WASDE	USDA WAOB	Monthly	~11th of month, 12:00 ET	C2–C6 — production and export forecast revisions
Weekly Export Sales Report	USDA FAS	Weekly	Thursday 8:30 AM ET	C1–C4 — China/Mexico/Japan demand signal
USDA Weekly Livestock Slaughter	USDA NASS	Weekly	Monday/Friday	C1–C2 — sow slaughter component for breeding herd signal
CFTC COT Report	CFTC	Weekly	Friday ~3:30 PM ET	Positioning / crowding — all months

Broiler Hatchery Report	USDA NASS	Weekly	Wednesday	Competing protein signal — indirect C2–C4 impact
USDA NASS Poultry Production	USDA NASS	Monthly	~last week of month	Competing protein structural signal
MSHMP Disease Monitoring	U of MN	Weekly	Online continuous	PRRS / disease risk — front spread tail event monitor
USDA APHIS Disease Alerts	USDA APHIS	As needed	Real-time	Tail event — PEDv, ASF, FMD detection

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Lean Hog is one of the most seasonally structured and biologically driven markets in the commodity universe. The forward curve encodes expectations about production cycles that are set by breeding decisions made months in advance, and demand patterns that repeat with a regularity driven by consumer behaviour and the calendar. These structural features make the seasonal baseline highly informative as a starting point — but it must always be adjusted for the current year's supply pipeline (as revealed by the Hogs & Pigs report), the trajectory of feed costs, the state of export demand (particularly from China and Mexico), and the ever-present risk of disease disruption.

The most reliable spread edges in Lean Hog come from correctly reading the Hogs & Pigs weight-distribution data (a 2–4 month supply pipeline view), tracking weekly slaughter vs prior year to identify whether the market is running tight or loose relative to seasonal expectations, monitoring sow slaughter for early signals of breeding herd changes, and maintaining active awareness of disease surveillance data — which can invalidate any fundamental view within days in a disease event scenario.